

REPORT
OF THE
STATE VETERINARIAN
OF
CALIFORNIA

For the Two Years Ending June 30, 1906

CHARLES KEANE, State Veterinarian



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REPORT OF THE STATE VETERINARIAN.

*To His Excellency, GEORGE C. PARDEE,
Governor of California.*

DEAR SIR: I respectfully beg leave to submit my report as State Veterinarian for the two years ending June 30, 1906.

During the two years just past our efforts to thoroughly organize the work of this department have been productive of good results. The Legislature of 1905, at my request, enacted an amendment to the County Government Act authorizing Boards of Supervisors, in their discretion, to appoint county inspectors to coöperate with the State Veterinarian in the control and eradication of infectious diseases of live stock. Practically all of the counties situated in the central and southern portions of the State complied with this Act, and they have also adopted uniform county live stock ordinances.

The Legislature of 1905 also provided this department with an assistant. The services of this assistant, Dr. W. E. D. Morrison, who is stationed at Los Angeles, have materially assisted the development of this work. New work has been taken up in counties where practically nothing had been done heretofore, especially in the work of eradicating the cattle tick. As the territory of this State is large, and the calls upon this office must be answered promptly, the appointment of Assistant State Veterinarian has been of material benefit to the live stock interests.

It is easy to underestimate the importance of this branch of the public service. But when a serious disease has obtained a foothold among the live stock of some locality, then the public realizes to what extent it is dependent upon our services.

In addition to the work of control and eradication of diseases of live stock, and making inspections for shipments in conformity with State and county regulations, this department has been instrumental in the organization of stock owners into associations for the purpose of coöperating with us in our line of work. Another association, composed of officials engaged in live stock sanitary work, was organized under the name of the California Live Stock Sanitary Association. These organizations have been productive of much good to the live stock interests of the State.

This department is of comparatively recent origin, the office of State Veterinarian having been created by the Legislature of 1899. While

the law creating the office provides, in a general way, for the duties of the State Veterinarian, his legal status in the work of eradicating certain infectious diseases is not specific enough; and for that reason I would respectfully recommend that the next Legislature grant us the necessary authority to eradicate infections where recalcitrant parties will not follow our instructions. For example, under the present law we have the authority to establish and maintain a quarantine on a ranch infested with cattle ticks (*Boophilus annulatus*), but our right to eradicate these ticks without the owner's coöperation is questionable. These cattle in question may be what we term immune cattle; that is, they may carry the fever ticks and will not contract Texas fever. The owner of the cattle may have no desire to move the cattle from his own ranch which is under quarantine, and consequently the established quarantine does not affect him. At the same time there is always the possibility of some of these cattle breaking through his fences and infecting other clean ranches.

TEXAS FEVER.

The most important work, from an economical standpoint, that this department is pursuing is the control and eradication of the cattle tick (*Boophilus annulatus*), the carrier and transmitter of Texas splenic, or southern cattle fever. During the past two years this work has been vigorously prosecuted in every infested county in the State.

During the present year all of Merced County and a portion of Madera County were placed above the State and Federal quarantine lines.

In that portion of Madera County which was not taken out of quarantine there was only one focus of tick infestation. This infested ranch consists of some 100,000 acres or more, and early in the present year all of the cattle and horses were taken off of the infested fields after the cattle had been dipped in oil in order to destroy the ticks remaining on them. After a sufficient length of time the ticks remaining on the fields would die from starvation. But unfortunately during early summer considerable of this property was flooded and cattle were again brought on to the infested fields, thereby setting our work back in that section for one season.

In Fresno County in the neighborhood of Laton and Riverdale one ranch on which the process of tick reclamation was progressing favorably was also flooded for a time, with the misfortune that the cattle from this infested ranch broke through the fences and infested several other small dairy ranches. However, this is now entirely under control, and reclamation work is being pursued on these as well as several other small foci of infestation in this district. The balance of Fresno County is free from ticks.

Kings County is now entirely free from Texas ticks. It is not advisable to remove this county from the quarantine area until the counties north of Kings are clean.

In Tulare County there are three ticky ranches, on which all necessary work is being done, with the prospects of having them clean this winter.

Kern County has also now only three small foci of infestation, on which we have been working during the past year.

In San Luis Obispo County, in that strip of territory west of the backbone of the Santa Lucia range of mountains, there is considerable tick infestation on the small dairy ranches. During the past two years many dipping plants have been built there, and tick eradication work is being vigorously prosecuted.

In Santa Barbara, Ventura, Los Angeles, San Bernardino, Orange, Riverside, and San Diego counties considerable tick infestation exists. While much improvement has been noted in some of these counties during the past two years, there is considerable work ahead to be done in this line of tick eradication.

Only one infraction of the State quarantine was called to the attention of this department during the two years just past. One of our large cattle outfits moved some cattle from Fresno County into Monterey County (Monterey County is above the State and Federal quarantine line) without first securing permission from this office. Complaint was sworn to by the State Veterinarian in Salinas, and after considerable delay the defendant plead guilty to the charge and was fined \$100 and costs. I have also urged the prosecution by county authorities of several individuals who have infringed county live stock ordinances, and convictions were secured in each case. This is very unpleasant work, but it becomes necessary from the fact that if our quarantines are not maintained the work retrogresses and stock interests suffer.

While maintaining a State quarantine against all cattle in the infested counties, we also establish quarantines on the individual ranches in the quarantined area, thereby not only is territory outside of the quarantined area protected, but the clean ranches in the quarantined area as well. By a strict maintenance of these quarantines losses from Texas fever in the quarantined area have been reduced greatly, and no outbreaks have occurred outside of the quarantined area.

During the last session of Congress the Bureau of Animal Industry, U. S. Department of Agriculture, received an appropriation of approximately \$80,000 for use in the eradication of cattle ticks in those states and territories that have adequate laws on the subject, and whose officials would coöperate with the Federal authorities in this work. I corresponded with the Chief of the Bureau of Animal Industry with

the end in view of obtaining a part of this assistance, and later received the following reply:

DR. CHARLES KEANE, *State Veterinarian,*
Sacramento, Cal.

SIR: This Bureau is in receipt of your letter of the 29th ultimo respecting the eradication of cattle ticks in California, and your purpose to apply to the Department for the release from quarantine of several counties in the San Joaquin Valley next year. It is noted you state that while the State of California has no law compelling the owners of infected ranches to clean same, yet the State authorities place restrictions on the movement of infected cattle, thereby stimulating the efforts of the owners to eradicate infection, and further, that the stockmen of that State are alive to the situation and will render all the assistance required. Under these conditions it appears that profitable work in tick eradication can be accomplished in your State during the present season; yet it is to be hoped that by another year adequate legislation will have been enacted and funds provided for the work in that State. The State laws should give to local officials authority to enter premises, to inspect live stock and enforce quarantine, which may include single farms or districts defined by county or other boundary lines, and to control the movement of live stock. Such officers should be empowered to enforce disinfection of animals and premises by the application of parasiticides or by other practicable means, such as changing animals from one inclosure, field, or premises to another, as may be necessary. The laws should also give authority to State officials to issue rules and regulations to establish and maintain quarantine, as above indicated, and also to confer authority upon Federal representatives to act as officials of the State in such matters. In this connection a general stock law is considered eminently important.

For the purpose of coöperating with your State in eradicating cattle ticks, the Bureau will detail Dr. Wm. M. Mackellar, inspector, to assume immediate charge for the Bureau of tick eradication in the counties which you desire to have released from quarantine the coming year. Dr. Mackellar will be instructed to confer with you at Sacramento regarding different phases of the work it will be practicable to do during the present season. Other inspectors will be assigned to duty in that State under Dr. Mackellar.

Very respectfully,

A. M. FARRINGTON,
Acting Chief of Bureau.

Subsequently nine Federal inspectors were detailed by the Bureau of Animal Industry to this State to coöperate with us in tick eradication work, and in the short period they have been here this year the results have been extremely gratifying. As the Department of Agriculture insists on adequate State laws and thorough coöperation, it will rest with the coming Legislature to give us this assistance in order that this work shall be continued next year. From the manner in which this work is now progressing it is only a matter of a few years when this State will be entirely free from cattle ticks.

ANTHRAX.

Several outbreaks of this disease have occurred during the past two years. This disease is mostly confined to the bottom lands along our large waterways. One severe outbreak took place on a farm close to the City of Napa during the fall of 1905. About one hundred head of

cattle and horses succumbed. When I visited this place I found that a number of the carcasses had been skinned and delivered to a nearby tannery. Other gross infractions of sanitary laws had taken place also. Skinned carcasses were dragged over the county roads and thrown into ditches, from which places there was great danger of the infection being carried to neighboring farms during the winter rains. I removed the hides from the tannery and ordered them cremated; the carcasses were also cremated. As anthrax is communicable to the human family even through the skin of an animal after it has been manufactured into leather, great care must be taken in the cremation of the entire carcass. When the carcasses of animals which have died from anthrax are left lying upon the ground the contagium will remain in the soil for years, and there is great danger of it being carried by animals or birds or by water to neighboring places. The destruction of all carcasses by cremation is the plan adopted by this department.

On several large cattle ranches where anthrax has been an annual visitor for years past, and where the losses have been extensive, it is now the plan to remove the cattle from the infected fields during the late summer and early fall. This disease usually makes its appearance at about the same time each year, and at this time the feed is usually well eaten off, so that the stockman suffers very little loss of forage from this procedure. When the cold weather comes on cattle can again be placed on these fields with comparative safety. Where this change of pasture can not be accomplished preventive vaccination is practiced, with a little success.

BLACK LEG.

Black leg, also termed black quarter and symptomatic anthrax, is a disease that attacks young cattle, and in former years was the cause of severe losses in this State. Veterinary science has developed a system of preventive vaccination against this disease, which, if properly and systematically performed, reduces fatalities to a minimum. The losses in this State are very light, as vaccination is universally performed by stock owners.

CONTAGIOUS ABORTION IN COWS.

During the summer of 1905 a severe outbreak of contagious abortion occurred among the dairy cows at the Napa State Hospital for the Insane. This disease had gained considerable headway when I visited the hospital, and before the trouble was checked the majority of the milch cows had aborted. While it is seldom that a fatality occurs from this disease, it has a decided economic importance from the fact that after abortion many of the cows dry up in their milk supply for an entire season. There is always a tendency left in these cows to abort again the next season. The disease is very pernicious in character, and

being infectious must be combated by scrupulous cleanliness of the dairy barn and surroundings, and liberal use of disinfectants.

HOG CHOLERA.

The losses from hog cholera have been greatly lessened during the past two years, few outbreaks having been reported. This is entirely due to the fact that hog-raisers recognize the infectious character of this disease, and consequently adopt the usual sanitary precautions to prevent the introduction of the infection into their herds.

SHEEP SCABIES.

The sheep industry in this State has experienced considerable prosperity during the past few years. Feeding conditions have almost universally been good, with a brisk demand and good prices for both mutton and wool. Still, the presence of scabies or mange in many of our flocks has reduced the profits that otherwise would have occurred if the sheep were clean. This affection is a serious menace to the sheep industry of this as well as other Western states. Sheep do not fatten so readily when affected with scab, and if the disease is advanced to any great extent it is sometimes the occasion of fatalities. It also affects the wool clip in both quantity and quality. The disease is easily eradicated from a flock by thorough and systematic dipping. A thorough State inspection and supervision of sheep shipments is an absolute necessity in order to completely eradicate this disease.

The sheep inspection law enacted by the Legislature of 1903 for the control and eradication of scabies of sheep has, in some localities, proved to be ineffective. Representative California sheepmen recognize the financial benefits that will accrue from the complete eradication of scab, but it is only by the enactment and enforcement of a good law that beneficial results will ensue.

Under the present system of inspection the deputy State sheep inspectors are appointed and removed by the various Boards of Supervisors. While the State Veterinarian is the State sheep inspector, he is given no supervision over his deputies. Some of the deputy inspectors have proven incompetent, while other inspectors have been known to have issued clean bills of health for sheep which were badly infected with scab. This has happened not only once, but many times. Acting under the present law on the subject the State sheep inspector (State Veterinarian) is powerless to rectify this.

As an example, the following are a number of shipments of sheep received at the Los Angeles and San Francisco markets, the majority of which shipments were certified to by deputy inspectors as clean sheep, when as a matter of fact they were infected with scabies to a

greater or less extent. The balance of the shipments were unaccompanied by certificates of any kind. The names of the consignees, consignors, and deputy sheep inspectors who certified the shipments are omitted in this report:

Date of Shipment.	Number of Sheep.	Station Loaded.	Destination.
Sept. 24, 1904	650	Santa Barbara Co.	Los Angeles
Nov. 5, 1904	475	Saticoy	Los Angeles
Nov. 17, 1904	378	Saticoy	Los Angeles
Dec. 19, 1904	503	Florence	Los Angeles
Jan. 31, 1905	290	Florence	Los Angeles
Apr. 25, 1905	150	Burbank	Los Angeles
May 10, 1905	505	San Diego	Los Angeles
May 16, 1905	492	Santa Monica	Los Angeles
May 17, 1905	800	San Bernardino	Los Angeles
May 17, 1905	600	San Diego	Los Angeles
May 18, 1905	392	San Diego	Los Angeles
May 25, 1905	693	Santa Barbara Co.	Los Angeles
June 2, 1905	345	Santa Barbara Co.	Los Angeles
June 6, 1905	448	Ventura	Los Angeles
June 10, 1905	1,012	San Bernardino Co.	Los Angeles
July 21, 1905	334	Santa Barbara Co.	Los Angeles
Aug. 15, 1905	120	Gilroy	San Francisco
Aug. 16, 1905	155	Corcoran	San Francisco
Aug. 21, 1905	160	Puente	Los Angeles
Aug. 21, 1905	1,137	Bay City	Los Angeles
Aug. 22, 1905	190	Ventura	Los Angeles
Sept. 6, 1905	310	Bethany	San Francisco
Sept. 13, 1905	480	Buttonwillow	San Francisco
Sept. 20, 1905	284	Bethany	San Francisco
Oct. 31, 1905	1,159	Saticoy	Los Angeles
Nov. 6, 1905	1,059	Bethany	San Francisco
Feb. 9, 1906	827	San Pedro	Los Angeles
Mar. 7, 1906	240	Connor	San Francisco
Mar. 26, 1906	264	Alhambra	Los Angeles
Apr. 10, 1906	1,280	Bakersfield	Los Angeles
Apr. 17, 1906	825	Bakersfield	Los Angeles
May 7, 1906	983	Mojave	Los Angeles
May 23, 1906	723	Dinuba	San Francisco

GLANDERS.

There is hardly a county in the State that has been entirely free from this disease for a number of years past. The State Veterinarian and his assistant have destroyed a large number of horses and mules affected with this malignant disease during the past two years. Many head of stock have also been destroyed by order of the various live stock inspectors and officials in charge of sanitary work in the larger cities. It is an unfortunate circumstance that upon information we will occasionally discover a glandered animal in the possession of a citizen who will endeavor to hide its presence from sanitary officials, understanding at the same time that this disease is communicable to man. During 1905 it became necessary to establish a temporary quarantine against the portion of Arizona in the neighborhood of Needles, San Bernardino County. A serious outbreak of glanders occurred in

Arizona Territory around this section, but it was quickly stamped out by the Territorial Veterinarian. Later our quarantine was raised.

Early during the present year it became necessary for me to prosecute an individual in Stockton. This party had in his possession an animal affected with glanders and would not submit to its destruction. The animal was later destroyed.

TUBERCULOSIS.

While this disease is prevalent on many of the dairy farms of this State, this department has made no systematic effort to combat it. The present laws in force and our lack of help and financial assistance preclude the taking up of a work of this magnitude. The scientific investigations in relation to this disease are receiving the attention of the most eminent scientists. Several European and American investigators have announced within the past few years that they have been successful in manufacturing a material which when injected into cattle would confer immunity against tuberculosis. The announcement of Von Behring, the discoverer of diphtheria antitoxin, that he had also discovered a material which would grant immunity to cattle against tuberculosis, received the widespread attention of scientists and sanitarians. If this proved to be true the eradication of tuberculosis in cattle would be a comparatively easy task in time. The reports of several European commissions appointed for the purpose of investigating Von Behring's bovine vaccine were very favorable. I received the permission from the Superintendent of the Napa State Hospital to test this bovine vaccine on some of the calves at the hospital dairy. Twenty calves were placed at my disposal early in 1905, into ten of which the material was injected according to the instructions of Von Behring; the balance of the calves were used as control animals. At the present time all of the injected calves are in excellent condition, but it is yet too soon to give definite results. This same procedure is being carried out by me at a dairy farm in the vicinity of Sacramento. It is to be hoped that these experiments will prove to be as successful as some of those reported by several European commissions.

INSPECTIONS.

Since January 5, 1905, 508 horses and mules, comprising sixty separate inspections, were made for shipments to other States and Territories in compliance with the regulations at points of destination.

Since January 5, 1905, 7,695 cattle, comprising sixty-eight separate inspections, were made from points in the modified quarantined area, and during the open season, to points outside of the quarantined area in this State.

Respectfully submitted.

CHARLES KEANE,
State Veterinarian.

